

Exogenous Incomplete Markets

Canonical Bond Economy Model

QEM – Macroeconomics III

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1 Setup

In the **exogenous incomplete markets** (bond economy) model, households are only allowed to trade a single one-period, non-state-contingent bond. There are no Arrow-Debreu securities, so markets are incomplete by assumption.

We assume that the aggregate endowment is constant over time, which implies that the equilibrium bond price q is also constant. The gross return on the bond is therefore constant and given by:

$$r = \frac{1}{q} - 1.$$

2 Budget Constraint

The period budget constraint for household i at history s^t is:

$$c_{i,t}(s^t) + a_{i,t+1}(s^t) = (1 + r) a_{i,t}(s^{t-1}) + y_{i,t}(s^t),$$

where:

- $c_{i,t}(s^t)$: consumption of household i at history s^t ,
- $a_{i,t+1}(s^t)$: gross savings (asset position) chosen at s^t , carried into next period,
- $y_{i,t}(s^t)$: idiosyncratic labor income at s^t .

Households face a **borrowing constraint**:

$$a_{i,t+1}(s^t) \geq -b, \quad \forall i, \forall s^t,$$

where $b \geq 0$ is the maximum amount that can be borrowed. This constraint is the same for all households.

Remark 1. Households are **ex-ante identical** (same preferences, same borrowing limit), but later differ due to different realizations of their idiosyncratic income shocks.

3 Implications of Incomplete Markets

The incomplete markets structure has several important implications:

- **Consumption is not perfectly smooth**, even if aggregate output Y_t is constant, because households cannot fully insure against idiosyncratic shocks.
- **Precautionary saving**: households save an additional amount above what the permanent income hypothesis would predict, precisely because future income is stochastic rather than deterministic.
- **Endogenous inequality** in both consumption and wealth arises from heterogeneous shock histories.
- **Mobility** within the consumption and wealth distributions occurs over time as shocks accumulate.

4 Household's Problem

All households solve the same infinite-horizon optimization problem, taking r as given:

$$\max_{\{c_{it}, a_{i,t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}),$$

where $\mathbb{E}_0$ denotes the expectation over future sequences of idiosyncratic income shocks, and $\beta \in (0, 1)$ is the discount factor.

Subject to:

$$c_{it} + a_{i,t+1} = (1 + r) a_{it} + y_{it}, \quad (1)$$

$$a_{i,t+1} \geq -b. \quad (2)$$

4.1 Modified Euler Equation

Let $\lambda_t \geq 0$ be the Lagrange multiplier on the borrowing constraint (2). The first-order condition yields the **modified Euler equation**:

$$u'(c_{i,t}) = \beta(1 + r) \mathbb{E}_t[u'(c_{i,t+1})] + \lambda_t.$$

Since $\lambda_t \geq 0$, we obtain the inequality:

$$u'(c_{i,t}) \geq \beta(1 + r) \mathbb{E}_t[u'(c_{i,t+1})].$$

Remark 2. *The multiplier $\lambda_t > 0$ when the borrowing constraint binds. This raises the marginal cost of consuming today relative to postponing consumption, thereby **incentivizing additional saving** beyond the standard Euler equation prescription.*

4.2 Bounded Wealth Accumulation

Wealth accumulation remains bounded if:

$$\beta(1 + r) < 1.$$

This condition ensures that the present value of the infinite income stream is finite and that the household does not wish to accumulate infinite wealth (see Ljungqvist & Sargent, Ch. 17).

5 Income Process

We assume that y_{it} follows a **first-order ergodic finite-state Markov chain**:

$$\Pr(y_{i,t+1} \mid y_{i,t}, y_{i,t-1}, \dots, y_{i,0}) = \Pr(y_{i,t+1} \mid y_{i,t}).$$

The transition probabilities are denoted:

$$\Pi(y'_i, y_i) = \Pr(y_{i,t+1} = y'_i \mid y_{i,t} = y_i),$$

and Π is the corresponding **transition matrix**. The entry $\Pi(y'_i, y_i)$ also gives the fraction of agents transitioning from state y_i to y'_i in the cross-sectional distribution.

Since the chain is **ergodic** (it settles into stable long-run behavior and “forgets” its starting point), it admits a **unique stationary distribution** $\Pi^*(y_i)$ satisfying:

$$\Pi^*(y'_i) = \sum_{y_i} \Pi(y'_i, y_i) \Pi^*(y_i).$$

6 Recursive Formulation

6.1 Bellman Equation

Dropping time subscripts and exploiting the Markov structure, the household's problem can be written recursively. The **value function** $V(a_i, y_i)$ satisfies:

$$V(a_i, y_i) = \max_{c_i, a'_i} \left\{ u(c_i) + \beta \sum_{y'_i} \pi(y'_i, y_i) V(a'_i, y'_i) \right\},$$

subject to:

$$\begin{aligned} c_i + a'_i &= (1 + r) a_i + y_i, \\ a'_i &\geq -b. \end{aligned}$$

The **individual state vector** is (a_i, y_i) , where:

- a_i is the individual asset holding (endogenous state variable),
- y_i is the current income realization (exogenous state variable).

6.2 State Space and Measure

Let μ be the **probability distribution** (measure) of agents over states (a, y) . We define:

- $A \equiv [-b, \bar{a}]$: the asset space (bounded below by $-b$, above by some $\bar{a}$),
- $\mathcal{Y}$: the finite set of possible income values y_i ,
- $S \equiv A \times \mathcal{Y}$: the **state space** for each individual agent.

Let $\mathcal{A} \subseteq A$ and $\mathcal{Y} \subseteq Y$ denote measurable subsets of the asset and income spaces, respectively.

6.3 Transition Function

The **conditional transition function** Q describes how the joint distribution over (a, y) evolves. Given current state (a_i, y_i) , the probability of transitioning into $\mathcal{A} \times \mathcal{Y}$ is:

$$Q(\mathcal{A} \times \mathcal{Y} \mid a_i, y_i) = \mathbf{1}[a'(a_i, y_i) \in \mathcal{A}] \sum_{y'_i \in \mathcal{Y}} \pi(y'_i, y_i),$$

where $a'(a_i, y_i)$ is the optimal savings policy function obtained from the Bellman equation, and $\mathbf{1}[\cdot]$ is the indicator function.

Remark 3. *The transition function Q is the key object linking individual optimal decisions to the evolution of the aggregate distribution μ . A **stationary equilibrium** requires μ to be a fixed point of the operator induced by Q .*